Demodulate, Then Attend: Exposing Amplitude Dynamics for Compact EEG Models

Anonymous CVPR submission

Abstract

Modern electroencephalography (EEG) models increasingly rely on larger networks and more pretraining. We revisit a simpler choice: the signal representation on which learning operates. We advocate demodulation before attention: extract each band's amplitude envelope and pair it with the corresponding waveform, exposing both amplitude dynamics and oscillatory morphology. Scattering theory motivates this separation and explains why temporal averaging can discard useful modulation structure. A simple tokenizer applies a learnable filterbank and embeds matched local waveform and envelope sequences into an electrode–band–time grid. A compact axial Transformer reads this representation. With the reader and training recipe fixed, this representation improves TUEV Cohen's κ from 0.576 to 0.711, an absolute gain of 0.135, while using fewer parameters; the same construction achieves 0.699 PR-AUC on CHB-MIT with 1.62 million parameters and no pretraining. With the filterbank and reader unchanged, waveform-only and envelope-only tokens reach 0.498 and 0.642 on TUEV, compared with 0.711 for the pair. These results motivate a representation-first approach to compact EEG models: make useful signal structure available in the tokens, then learn how it is organized across time, electrodes, and frequency.

1. Introduction

EEG modeling has progressed from compact convolutional networks to Transformers pretrained on large collections of recordings [6, 7, 15, 17]. Yet larger models do not consistently outperform compact supervised specialists [8]. This observation motivates a more basic question: what representation should an EEG model learn from? We examine the signal computations performed before attention and the signal properties they expose.

Clinical EEG descriptions characterize morphology, recurrence, evolution, and spatial distribution [5]. A sharp discharge and a smooth oscillation differ in waveform shape; sustained activity and separated bursts differ in how their strength evolves. These properties motivate two descriptions of each frequency band. The signed *waveform* retains oscillatory shape, while the amplitude *envelope* tracks how activity strengthens and weakens. Their correspondence matters: a transient has a local shape and occurs within an evolving pattern of activity.

These two descriptions suggest a concrete input choice. Raw-patch embedding leaves the envelope implicit: exposing bandwise amplitude requires filtering followed by a nonlinear modulus, which a linear patch projection cannot compute. Supplying an envelope performs this operation before learning, but removes polarity and phase detail. We therefore retain the corresponding signed waveform and ask whether the two views are more useful together than either alone. This leads to *demodulation before attention*: expose amplitude dynamics without replacing the waveform that carries them. We hypothesize that this input choice simplifies a compact encoder's learning problem.

Scattering theory clarifies what should be preserved after demodulation. Bandpass transforms and modulus operations expose subband amplitude, but subsequent averaging can discard temporal modulation structure [1]. An envelope trajectory can distinguish burst patterns with similar average band energy. At the same time, taking a modulus discards waveform polarity and phase detail. These observations motivate preserving the envelope sequence together with its signed waveform: demodulation exposes a useful view of the signal without making that view the entire representation.

The resulting tokenizer is simple. A learnable filterbank decomposes each channel into bands, and an analytic transform extracts their envelopes before patching. Shared projections pair waveform and log-envelope sequences from the same electrode, band, and interval. The tokens form an electrode–band–time grid, read by an ordinary axial Transformer. Attention relates intervals along time, locations across channels, and components across bands. The complete model is trained with supervised classification alone.

Frequency-aware frontends provide important precedents. CBraMod adds patchwise spectral and convolutional waveform features [17]. TFM-Tokenizer learns discrete motifs from waveform and time–frequency representations [12]. The Morlet Spectral Transformer preserves

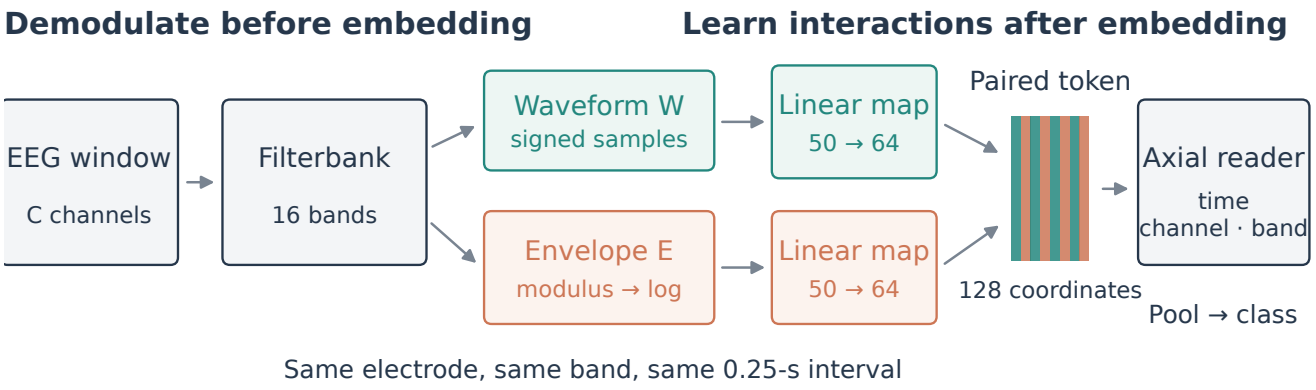


Figure 1. **Demodulate, then attend.** A learned filterbank produces band-limited waveforms. The analytic modulus supplies a corresponding amplitude envelope. Shared projections embed both 50-sample sequences into one token, retaining their electrode, band, and time indices. Six axial Transformer blocks learn interactions on this grid. The analytic transform is applied to the full input window before patching.

time-resolved wavelet amplitudes [13]. We study a specific continuous input representation: the local pairing of a band's waveform and envelope trajectory before attention. Holding the reader fixed makes it possible to compare this explicit signal representation with raw learned rows, and to separate the contributions of its two content views through controlled replacements.

The reference reaches $\kappa = 0.711$ on TUEV and PR-AUC 0.699 on CHB-MIT using 1.62 million parameters. On TUEV, replacing it with raw patches projected into the same number of learned rows reduces κ to 0.576 under the same reader, split, and training recipe: an absolute difference of 0.135 with fewer parameters in the proposed model. Holding the filterbank fixed further separates the two content views: waveform-only and envelope-only tokens reach 0.498 and 0.642, respectively, while their pairing reaches 0.711. F1 improves in all six event categories in the raw-row comparison. A separate linear readout of frozen tokens checks that both intended trajectories remain available before attention.

We propose demodulation before attention as a simple principle for EEG tokenization, implement it with local waveform–envelope pairs, and examine its effect with a fixed compact reader. The resulting model connects an explicit signal representation to clinical prediction without a separate pretraining stage. More broadly, this work asks which useful signal computations should precede learning, so that an encoder can build on directly available structure in the recording.

2. Related Work

2.1. EEG pretraining and tokenization

EEG foundation models address transfer across subjects, tasks, and recording systems. BIOT organizes channelwise signal segments into a common sequence [18]. LaBraM learns discrete neural codes through spectrum prediction [6], and EEGPT uses masked representation learning with alignment [16]. CBraMod separates spatial and temporal interactions in a criss-cross Transformer [17]; REVE extends large-scale pretraining to heterogeneous acquisition setups [3]. These methods combine input representations, learning objectives, and encoder designs to obtain transferable features.

TFM-Tokenizer instead makes the frontend a primary object of study: waveform and spectral paths produce a vocabulary of time–frequency motifs that can serve different encoders [12]. Our construction uses continuous, explicitly demodulated band content. It does not require a learned codebook or a separate tokenizer pretraining stage. The question is which local signal views should remain available inside each token before the encoder mixes information across locations.

2.2. Amplitude-based EEG representations

Explicit amplitude information is well established in EEG decoding. The shallow network of Schirrmeister et al. uses squaring, pooling, and logarithmic compression to extract power-related features [15]. FBCNet combines a filterbank, spatial filtering, and variance aggregation [9]. EEGNet provides another compact architecture through factorized temporal and spatial filtering [7]. HTNet computes filter–Hilbert power at learned frequencies before neural decoding [11]. Hilbert envelopes and temporal fine structure have also been concatenated for EEG speech decoding [2].

Recent frontends also preserve spectral or amplitude variation. CBraMod projects a one-second patch spectrum and combines it with waveform features [17]. The Morlet Spectral Transformer uses wavelet modulus, temporal pooling, and frequency-specific spatial projections [13]. Our

distinction is narrower than introducing amplitude: a token embeds a sequence of envelope samples and the corresponding signed waveform from one band, while electrodes remain a token axis. This makes local content and the granularity of its representation explicit experimental variables.

2.3. Demodulation before learned processing

The deep scattering spectrum extracts subband envelopes with analytic wavelets and modulus operations, then recovers modulation structure that would be lost through averaging [1]. We adopt the demodulation principle while passing local trajectories to learned attention. The implementation uses trainable sinc filters [14], rather than a fixed scattering cascade. Its contribution is the token interface, not a new demodulation theorem.

3. Method

Let $x_c[t]$ be the normalized EEG signal at channel c. We construct one token per channel c, band b, and time interval p. The method has three steps: demodulate the bands, embed paired local content, and read the resulting grid (Fig. 1).

3.1. Bandwise demodulation

A shared bank of B learnable bandpass filters produces signed waveforms $w_{c,b}$ and their analytic envelopes:

$$\begin{aligned} w_{c,b} &= g_b * x_c, \\ e_{c,b} &= \left| w_{c,b} + \mathrm{i}\mathcal{H}[w_{c,b}] \right|. \end{aligned} \tag{1}$$

where $\mathcal{H}$ is the Hilbert transform. Each sinc filter g_b is parameterized by two cutoffs [14]. We use 16 filters, initialized with logarithmically spaced edges from 0.5 to 75 Hz, and a 201-sample impulse response at 200 Hz. The cutoffs are learned during supervised training.

For a narrowband amplitude-modulated component $w(t) = A(t)\cos\phi(t)$ with suitable separation between modulation and carrier frequencies, the analytic envelope approximates $A(t)$. Thus the modulus exposes amplitude dynamics that would otherwise need to be derived from oscillating samples. With general EEG, it is the well-defined amplitude of the filtered analytic signal; it need not identify a single physiological oscillator. We retain the signed waveform in parallel because taking a modulus removes its phase and polarity information.

Filtering and the analytic transform operate on the entire input window. Only afterward do we divide the two sequences into tokens. A 0.25-s token therefore summarizes a local part of an envelope estimated with context beyond that token. The token interval is not the filter's support and does not impose a 0.25-s Hilbert-estimation window.

3.2. Pair local waveform and envelope trajectories

For interval p of length L, let $\mathbf{w}_{c,b,p} \in \mathbb{R}^L$ and $\mathbf{e}_{c,b,p} \in \mathbb{R}^L$ contain corresponding waveform and envelope samples. We form

$$\mathbf{h}_{c,b,p} = \left[P_E \log(1 + \mathbf{e}_{c,b,p}) + \mathbf{b}_E \,;\, P_W \mathbf{w}_{c,b,p} \right], \tag{2}$$

with $P_E, P_W \in \mathbb{R}^{(d/2)\times L}$. Both projections are shared across channels, bands, and intervals. The implementation interleaves the two sets of coordinates, which is a fixed permutation of the concatenation in Eq. (2). We use $L = 50$ and $d = 128$. The logarithm compresses envelope dynamic range without averaging its trajectory.

The pairing is deliberately local. Each token places waveform and amplitude dynamics from the same signal component in a common learned representation. The encoder can then model their relationship without first matching independently aggregated views. We preserve all 50 input positions through separate projections rather than substituting a single power or variance statistic. There is no phase rotation or additional learned fusion module at this stage.

This construction also separates *available information* from *explicit computation*. A raw-patch projection can preserve all its input samples when its matrix has sufficient rank. It nevertheless does not output an amplitude envelope: the modulus is nonlinear. Our design supplies this computation before attention and retains the waveform from which it was derived. It introduces a representational bias, not an additional measurement of the underlying EEG.

3.3. A compact reader for the token grid

Tokens form a $C \times B \times P$ grid. Channel embeddings derived from electrode coordinates and learned band-index embeddings identify the two non-temporal axes. Each pre-normalized Transformer block applies self-attention along time, channels, and bands in sequence, followed by a feed-forward network. Time attention uses rotary positions. Each axis uses ordinary four-head attention; the reference has six blocks, width 128, and a feed-forward expansion of two.

The axes follow the representation: time relates intervals, channels relate recording locations, and bands relate signal components. Factorization permits this grid to be read without global attention over all its tokens. Global mean pooling, layer normalization, and a linear classifier predict the window label. The complete reference contains approximately 1.62 million parameters. We keep this reader fixed in the content experiments; it is not a separate proposed attention mechanism.

3.4. Checking the token's content

Before attributing a classification result to preserved content, we check that the two intended views can be recovered from the token. We freeze the trained frontend and fit separate linear readouts to waveform and log-envelope samples,

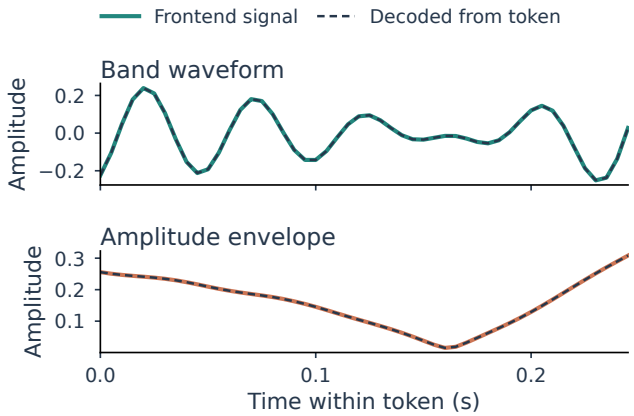


Figure 2. **What remains in a token?** A linear readout of frozen TUEV tokens recovers the waveform and envelope of a held-out band segment. Curves overlap at this scale. This checks local content preservation before attention; classification controls test utility.

Table 1. **Content recovery before attention.** Held-out normalized RMSE of linear readouts fitted to frozen tokens. Targets are the frontend's waveform and log-envelope samples.

Target	TUEV	CHB-MIT
Waveform	3.21×10^{-4}	1.33×10^{-4}
Log-envelope	4.13×10^{-7}	5.38×10^{-7}

using training windows only. Readout regularization is selected on validation windows; reconstruction is measured on held-out test windows. No classification labels enter this check, and no classifier weights are updated.

Both content views are recovered with small error (Tab. 1 and Fig. 2). The two 64×50 projection matrices have full column rank in both trained models, explaining why linear recovery is possible. This is a functional check of the representation, not evidence that every retained coordinate is needed for classification. In particular, reconstruction does not establish whether a waveform-only or envelope-only token would perform equally well. Those questions require the content controls in Sec. 5.2.

4. Experimental Setup

Tasks and data. We study six-class event classification on TUEV [10] and binary seizure-window classification on CHB-MIT [4]. Both use 16 bipolar channels at 200 Hz. TUEV contains 5-s event-centered windows, with subject-separated training and validation sets drawn from the official training partition and the official evaluation subjects reserved for testing. CHB-MIT uses 10-s windows and case-separated partitions. Its evaluated protocol includes additional seizure-centered windows in each partition; the test set has 17,386 windows, including 276 positives. All within-dataset comparisons must use these same windows. Appendix A gives normalization, partition counts, and the exact sampling protocol.

Training. The reference and content controls are trained from scratch with AdamW, a learning rate of 10^{-4}, weight decay 0.05, cosine scheduling, batch size 32, and a maximum of 20 epochs. The configured dropout is 0.35. TUEV uses cross-entropy with label smoothing 0.1; CHB-MIT uses focal loss with $\alpha = 0.25$ and $\gamma = 2$. All content arms share the same data augmentation, optimizer, reader, and checkpoint-selection rule. Appendix B details the transformations and validation schedule.

Evaluation. We select checkpoints by validation κ on TUEV and validation PR-AUC on CHB-MIT. The held-out test set is used for reporting. TUEV reports κ, balanced accuracy, and support-weighted F1. CHB-MIT reports trapezoidal PR-AUC, AUROC, and balanced accuracy at threshold 0.5; we additionally give average precision. The reference and content controls use seed 2. Historical baseline cells summarize seeds 0, 1, and 2 with sample SD, except the available shared-protocol TUEV LaBraM and EEGPT runs, which use seed 0. Baseline recipes and task adaptations are documented in Appendix D. Empty cells have no compatible completed run; scores from other montages or window sets are not substituted.

5. Results

5.1. Clinical performance

The 1.62M reference achieves $\kappa = 0.711$ on TUEV and PR-AUC 0.699 on CHB-MIT (Tab. 2). It exceeds the highest recorded pretrained baseline primary scores, from REVE and TFM-Tokenizer respectively, by 0.027 and 0.073. CHB-MIT average precision is 0.702, close to its trapezoidal PR-AUC of 0.699.

To examine the frontend directly, we hold the reader and training recipe fixed and change its input (Tab. 3). A linear map turns each raw 50-sample channel patch into 16 learned rows of width 128, matching the reference token grid. Replacing these rows with bandwise waveform–envelope pairs improves all three metrics, including an absolute κ gain of 0.135, with fewer parameters. The comparison tests the complete frontend; the content controls below separate its views. On CHB-MIT, paired input raises PR-AUC from 0.642 to 0.699, a gain of 0.058. The raw-row model has higher AUROC (0.928 versus 0.918), so this comparison does not imply improvement in every operating characteristic (Appendix B).

Table 2. **Clinical evaluation on shared data protocols.** Baselines reuse completed runs with their recorded task adaptations (Appendix D). Values with dispersion are mean ± sample SD; single-run entries have no SD. Size gives recorded parameter counts in millions for TUEV / CHB-MIT; unrecorded counts are shown as –. Bold marks the highest available score in each column.

		TUEV			CHB-MIT		
Model	Size (M)	κ ↑	Bal. acc.↑	wF1↑	PR-AUC↑	AUROC↑	Bal. acc.↑
Pretrained							
BIOT [18]	3.19 / 3.19	0.530±0.028	0.489±0.030	0.753±0.016	0.560±0.077	0.874±0.004	0.619±0.045
LaBraM [6]	– / 5.82	0.602	**0.655**	0.797	0.452±0.097	0.871±0.062	0.586±0.047
EEGPT [16]	– / 25.69	0.634	0.594	0.812	0.281±0.133	0.857±0.038	0.547±0.063
CBraMod [17]	17.85 / 30.65	0.645±0.022	0.603±0.011	0.813±0.012	0.233±0.098	0.783±0.047	0.545±0.020
REVE [3]	69.48 / 69.46	0.685±0.039	0.648±0.018	0.838±0.019	0.580±0.196	0.898±0.064	0.632±0.068
TFM-Tokenizer [12]	1.89 / 1.89	0.652±0.021	0.596±0.036	0.817±0.010	0.627±0.025	**0.929±0.006**	**0.634±0.016**
From scratch							
BIOT [18]	3.19 / 3.19	0.532±0.025	0.535±0.062	0.761±0.013	0.566±0.072	0.875±0.019	0.624±0.071
LaBraM [6]	– / 5.82	0.569	0.380	0.766	0.360±0.050	0.823±0.030	0.585±0.065
EEGPT [16]	– / –	0.578	0.530	0.783			
CBraMod [17]	17.85 / 30.65	0.564±0.024	0.574±0.010	0.767±0.004	0.317±0.205	0.780±0.133	0.602±0.088
REVE [3]	69.48 / –	0.319±0.014	0.370±0.042	0.655±0.004			
Demodulation-first (ours)	1.62 / 1.62	**0.711**	0.634	**0.849**	**0.699**	0.918	0.630

Table 3. **Changing the input, keeping the reader.** TUEV with the same six-block, width-128 axial Transformer and training recipe. Changes are computed before rounding.

Token input	Params (M)	κ	Bal. acc.	wF1
Raw patches	1.72	0.576	0.552	0.782
W+E pairs	1.62	**0.711**	**0.634**	**0.849**
Change (W+E − raw)		+0.135	+0.081	+0.067

Table 4. **Content controls with a fixed reader.** All bandwise variants keep 16 bands, two 64-coordinate projection blocks, and the same seed-2 training recipe. Repeating one view in both blocks preserves projection dimensions and parameter count. Cells without completed tests remain empty.

Token input	TUEV κ	CHB-MIT PR-AUC
Raw learned rows	0.576	0.642
Waveform only	0.498	
Envelope only	0.642	
Envelope + waveform	**0.711**	**0.699**

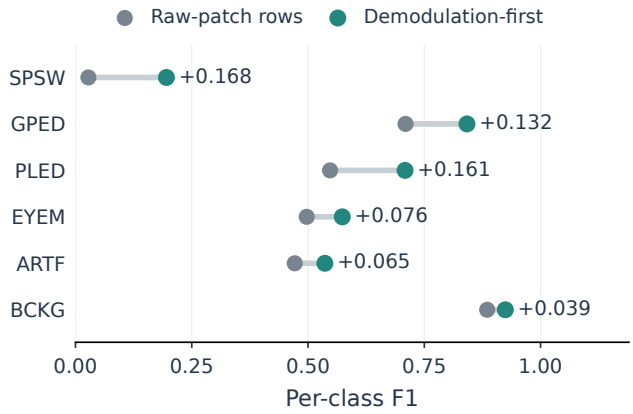


Figure 3. **TUEV gains across event categories.** Paired markers compare held-out F1 for raw learned rows and demodulation-first tokens with the same reader. Text gives the absolute difference. All six categories improve; the figure does not attribute these gains to the envelope alone.

5.2. Which content contributes?

We hold the filterbank, reader and training recipe fixed. Waveform-only and envelope-only arms fill both projection blocks with the same view, retaining the reference's parameter count and initialization (Tab. 4).

On TUEV, envelope-only tokens reach $\kappa = 0.642$, compared with 0.498 for waveform-only tokens. Pairing the views reaches 0.711: an increase of 0.213 over waveform alone and 0.070 over envelope alone. Balanced accuracy and weighted F1 follow the same ordering (Appendix B). Under this controlled replacement, explicit amplitude helps beyond filtered waveform, and retaining the waveform adds further benefit.

5.3. Where does event recognition improve?

The aggregate gain extends beyond the majority background category (Fig. 3). GPED F1 increases from 0.710 to 0.842, and PLED F1 from 0.547 to 0.709. F1 also improves for spike/sharp-wave events, eye movements, artifacts, and background. These results connect the overall score to both

recurrent-discharge and other event categories, rather than supporting an explanation confined to a single label type.

The class breakdown also shows what the aggregate metrics hide. Spike/sharp-wave F1 rises from 0.028 to 0.196, but recall remains 0.109. For eye movements, F1 improves despite a decline in recall, because precision rises. Table 6 in the appendix reports precision and recall alongside F1. This distinguishes improved category-level discrimination from uniform improvements in every operating characteristic.

6. Discussion and Conclusion

Scope. The matched comparison uses one reader, one seed, and one training recipe. CHB-MIT results describe the specified enriched window protocol rather than continuous-recording alarm performance. The filterbank and full-window analytic transform can introduce boundary and overlap effects; their implementation is documented in the appendix.

Conclusion. We construct EEG tokens by pairing local waveform and demodulated amplitude trajectories before attention. With the reader fixed, this input improves clinical prediction over raw learned rows; matched bandwise controls show an advantage over either content view alone. The result points to a simple design principle: expose useful signal computations without discarding complementary detail, then let attention learn their organization.

LLM Usage Statement

OpenAI Codex assisted with literature retrieval, manuscript drafting and editing, code and result-record inspection, figure preparation, and LaTeX compilation. It also assisted with numerical consistency checks and the organization of the experimental evidence.

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Table 5. **Evaluated window counts.** Counts include the specified CHB-MIT seizure-window enrichment.

Dataset	Train	Validation	Test
TUEV	68,445	15,487	29,421
CHB-MIT	316,205	21,184	17,386

A. Data and Evaluation Protocol

TUEV. The evaluated TUEV v2.0.1 cache contains five-second windows around annotated events. Signals are resampled to 200 Hz, bandpass filtered from 0.3 to 75 Hz, and notch filtered at 60 Hz. Values in microvolts are divided by 100 before storage. The official training subjects are sorted and split 80:20 into training and validation partitions; official evaluation subjects form the test partition. There are 232, 58, and 80 subjects, respectively. We do not apply a second scaling operation in the model loader.

CHB-MIT. The evaluated CHB-MIT v1.0.0 cache resamples the recordings from 256 to 200 Hz and extracts non-overlapping ten-second windows, dropping incomplete tails. Any overlap with an annotated seizure gives a positive label. Additional windows are extracted around seizure intervals with one-second pre/post margins and a five-second stride; these are labeled positive. This enrichment applies to training, validation, and test partitions, adding 1,781, 90, and 162 windows, respectively. It is part of the evaluated protocol, not training augmentation. A natural-grid-only evaluation would be a different test set.

Each window and channel is divided by its own 95th percentile absolute amplitude plus 10^{-8}. Thus amplitude in this experiment refers to variation within this normalization, not calibrated clinical voltage. Training uses cases chb01–chb19 and chb21; validation uses chb20 and chb22; testing uses chb23 and chb24. The repeated-subject recordings chb01 and chb21 both remain in training. The test set has 17,110 negative and 276 positive windows (1.59% positive). PR-AUC should be compared only on this same case and window composition.

Common montage. Both tasks use the following bipolar chains: FP1–F7–T7–P7–O1, FP2–F8–T8–P8–O2, FP1–F3–C3–P3–O1, and FP2–F4–C4–P4–O2. Adjacent pairs in each chain form the 16 channels. Older T3/T4/T5/T6 aliases are mapped to T7/T8/P7/P8. This is a 16-channel protocol; results from a 23-channel native pipeline require a separate evaluation rather than direct insertion into the main table.

Metrics. For multiclass TUEV, balanced accuracy is mean class recall and weighted F1 is weighted by test-class support. For binary CHB-MIT, PR-AUC is trapezoidal integration of the precision–recall curve. Average precision is also reported because it uses a different interpolation convention. The binary decision threshold is 0.5. Neither the threshold nor the checkpoint is selected on test labels.

B. Implementation and Training

Frontend. The 16 sinc filters share their parameters across channels. Each 201-sample symmetric FIR uses a Hamming window and zero padding of 100 samples on each side. The FFT-based Hilbert transform doubles positive-frequency coefficients, preserves the DC and Nyquist terms, and zeros negative frequencies. It is applied to the complete 5-s or 10-s filtered window. No extension beyond that window is used for the analytic transform. Responses near the boundaries may therefore depend on the finite-window convention.

The initialization is logarithmic; the learned bank is not constrained to retain constant-Q spacing or non-overlapping passbands. A token indexes a particular learned filter, rather than presupposing a physiological frequency category. Each content projection maps 50 samples to 64 coordinates. The envelope projection is affine and has a learned coordinatewise scale, absorbed into P_E and $\mathbf{b}_E$ in Eq. (2). The waveform projection has no bias and is initialized to zero. Its full trained rank in Tab. 8 confirms that it does not remain the initial zero map. Content controls preserve these projection shapes and initialization rules.

Reader. Bipolar channel coordinates concatenate the two electrode endpoints before a two-layer MLP. Dropout acts in the feed-forward sublayer. Global mean pooling precedes layer normalization and the classifier.

For C channels, B bands, and P intervals, the number of attention scores per head and block is

$$CBP^2 + BPC^2 + CPB^2 = CBP(P + C + B). \quad (3)$$

This describes attention-score complexity, not measured throughput. Projection and feed-forward costs remain proportional to token count and feature dimension. TUEV has $16 \times 16 \times 20 = 5{,}120$ tokens; CHB-MIT has $16 \times 16 \times 40 = 10{,}240$.

Training augmentation. One of five transformations is sampled uniformly per training batch: time reversal with probability 0.5, Gaussian noise with standard deviation 0.05 in stored signal units, masking 20% of temporal samples, masking 20% of channels, or masking Fourier bins with probability 0.2. Transformations act on the input before either frontend. The raw-row and band-content arms use the same choices. No training augmentation is applied during validation, testing, or token recovery.

Table 6. **TUEV test-category breakdown.** The two validation-selected checkpoints are evaluated on the same 29,421 windows. Bold marks the better tokenizer within each class and metric.

		Raw-patch rows			Demodulation-first		
Category	Support	Precision	Recall	F1	Precision	Recall	F1
SPSW	567	0.024	0.034	0.028	**0.925**	**0.109**	**0.196**
GPED	4,677	0.743	0.679	0.710	**0.915**	**0.779**	**0.842**
PLED	1,998	**0.661**	0.467	0.547	0.641	**0.792**	**0.709**
EYEM	329	0.396	**0.669**	0.497	**0.585**	0.562	**0.574**
ARTF	2,204	0.390	0.595	0.471	**0.463**	**0.637**	**0.536**
BCKG	19,646	0.900	0.871	0.885	**0.926**	**0.922**	**0.924**

Checkpoint selection. TUEV is validated once per epoch. CHB-MIT is validated every 200 optimizer steps and at epoch end. The patience counter counts non-improving validation evaluations; the stopping condition is checked at epoch boundaries. Patience is 20 for both tasks. The completed TUEV models finish all 20 epochs, whereas the CHB-MIT reference stops by patience after four epochs. Neither reported reference is terminated by its wall-clock budget. The selected TUEV reference checkpoint comes from evaluation 1 (zero-based), and the raw-row checkpoint from evaluation 14; the CHB-MIT reference comes from evaluation 130. The new TUEV waveform-only and envelope-only controls also select evaluation 1 and complete all 20 epochs. The CHB-MIT raw-row control selects evaluation 246 and stops by patience after six epochs. All selection uses the validation metric specified in Sec. 4.

Exploratory representation controls. The mean-envelope control replaces each 50-sample log1p-envelope patch by its within-patch mean, repeated 50 times. Filtering, the full-window Hilbert transform, both projection matrices, their initialization, and the waveform input remain unchanged. The bandmix control multiplies the 16-band token axis by a fixed dense orthogonal matrix obtained by QR factorization of a Gaussian matrix with seed 2. The same matrix acts on all channels, intervals, features and examples, before positional embeddings. It adds no trainable parameters. Each control changes one model configuration field relative to the paired reference. The mean-envelope arm reaches $\kappa = 0.695$, retaining amplitude changes between token intervals while averaging variation within each interval. The mixed-basis arm reaches 0.585. These supplementary comparisons are reported in Tab. 7; the main content comparison holds the physical grid fixed.

Complete content-control metrics. Table 7 reports all three task metrics for the completed comparisons. Each row uses the test predictions of its validation-selected checkpoint. The CHB-MIT raw-row and paired models have average precision 0.642 and 0.702, respectively. Original prediction and checkpoint hashes are retained with the result records.

Table 7. **Complete metrics for the content comparisons.** Seed 2 throughout. The two CHB-MIT rows use the same enriched 17,386-window test protocol. Bold denotes the best value per panel.

TUEV input	κ	Bal. acc.	wF1
Waveform only	0.498	0.510	0.735
Envelope only	0.642	0.577	0.812
Envelope + waveform	**0.711**	**0.634**	**0.849**
Waveform + mean E	0.695	0.612	0.842
Mixed band basis	0.585	0.530	0.788
CHB-MIT input	**PR-AUC**	**AUROC**	**Bal. acc.**
Raw learned rows	0.642	**0.928**	0.538
Envelope + waveform	**0.699**	0.918	**0.630**

C. Recoverability of Paired Content

Why reconstruction is possible. Write a token before positional embeddings as $\mathbf{h} = [P_E\boldsymbol{\ell} + \mathbf{b}_E; P_W\mathbf{w}]$, where $\boldsymbol{\ell} = \log(1 + \mathbf{e})$. If both projection matrices have column rank L, their Moore–Penrose inverses recover the two local input vectors in exact arithmetic:

$$\boldsymbol{\ell} = P_E^{\dagger}(\mathbf{h}_E - \mathbf{b}_E), \qquad \mathbf{w} = P_W^{\dagger}\mathbf{h}_W. \tag{4}$$

Interleaving token coordinates merely changes how the two blocks are indexed. The envelope follows by applying $\exp(\boldsymbol{\ell}) - 1$. This statement concerns the band-limited waveform and its envelope; it is not a claim that the complete original recording can be inverted from the filterbank.

Readout protocol. For each dataset we select 128 training, 32 validation, and 64 test windows with fixed seed 2. Four evenly spaced channel indices, all 16 bands, and three fixed interior patch indices are used per window. The

Table 8. **Trained projection matrices.** Each matrix has shape 64×50. Full column rank explains local recoverability; conditioning determines sensitivity to numerical error.

Dataset	Projection	Rank	Condition no.
TUEV	Waveform	50	3.90×10^4
	Log-envelope	50	11.66
CHB-MIT	Waveform	50	3.25×10^4
	Log-envelope	50	11.64

sample counts for fitting, validation, and testing are consequently 24,576, 6,144, and 12,288 tokens. Separate ridge readouts predict the 50 waveform or log-envelope samples from the 128 token coordinates. Each readout is shared across channel, band, and patch indices. The regularization coefficient is selected from $\{10^{-8}, 10^{-5}, 10^{-2}, 1\}$ by validation reconstruction error; both datasets select 10^{-8} for both targets.

Token coordinates are standardized with training-set means and standard deviations before fitting each ridge readout. Normalized RMSE is the square root of the total squared prediction error divided by the target's summed squared deviations from its per-coordinate test-set mean. For the physical envelope, obtained by exponentiating the log-envelope prediction, normalized RMSE is 1.12×10^{-6} on TUEV and 1.38×10^{-6} on CHB-MIT. A second check using the known projection weights and Eq. (4) produces the same qualitative result. Figure 2 shows a fixed example, not a reconstruction chosen to minimize error: the first held-out window and selected channel, band 12, and the middle of the three sampled intervals.

What a linear patch map cannot compute. For a nonzero filtered signal w, its envelope satisfies $|\mathcal{A}(-w)| = |\mathcal{A}(w)|$, where $\mathcal{A}$ is the linear analytic-signal operator. A linear map instead satisfies $F(-w) = -F(w)$. It therefore cannot equal the nonzero envelope map for both w and $-w$. This does not imply that a linear token loses the waveform, or that a nonlinear encoder cannot learn to compute an envelope. It identifies the computation explicitly moved ahead of the encoder in our representation.

Event categories. SPSW denotes spike and sharp waves, GPED generalized periodic epileptiform discharges, PLED periodic lateralized epileptiform discharges, EYEM eye movement, ARTF artifact, and BCKG background.

D. Historical Baseline Configurations

Runs and initialization. The main table reuses complete baseline result records on the shared 16-channel protocols. Each multi-run cell includes all three recorded seeds from one training recipe. The TUEV LaBraM and EEGPT entries use single completed runs through the authors' shared16 entrypoints, with scratch and pretrained initialization reported separately. The older LaBraM results on 26,067 evaluation windows are outside this comparison. CHB-MIT scratch EEGPT and REVE have no completed result on the evaluated 17,386-window protocol, so those cells remain empty. TFM's main-table entry uses both its pretrained tokenizer and pretrained encoder.

Task adaptations. Historical baselines retain their recorded, model-specific recipes. BIOT uses its channel-wise spectral input. The historical CBraMod implementation uses an 800-unit local classification head; its binary variant uses two logits and focal loss. CHB-MIT EEGPT uses a decoder derived from its sleep-stage configuration and a 16-to-19 channel mapping. The recorded REVE implementation uses direct fine-tuning with AdamW and omits the native learned-query context token. These scores therefore describe the documented task adaptations rather than exact reproductions of every model's published training pipeline. Their optimizer, loss, batch size, and training budget are preserved in the accompanying result records. The matched-reader comparison in Tab. 3 instead uses the same recipe in both arms.

Metrics and model size. The shared evaluation uses trapezoidal PR-AUC and a fixed binary threshold of 0.5. The native LaBraM keys `balanced_accuracy` and `f1_weighted` map to balanced accuracy and weighted F1. Model sizes include task-specific heads and can differ between datasets: for example, the recorded CBraMod instances contain 17.85M parameters on TUEV and 30.65M on CHB-MIT. Total parameter counts were not retained for the native shared16 LaBraM and EEGPT runs; the table leaves those sizes unreported rather than borrowing counts from different heads.